Learning & Teaching

Multiagent Systems Algorithmic, Game Theoretic and Logical Foundations

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Learning

The goal is to design an agent that learns to function successfully in an environment that is unknown and potentially also changes as the agent is learning.

In multiagent systems, the environment contains the other agents.

- Changes due to other agents' learning
- Changes due to own agent (protagonist)

Learning and Teaching

- Simultaneous learning, implies
  - *Dynamic system*
  - *Complex behavior (even with simple learning rules)*
- Teaching:
  - An agent learns based on what others have done in the past,
  - An agent acts to influence others

Learning and Teaching

Example: In stackelberg game player 1 has the incentive to teach player 2

	L	R
T	$1, 0$	$3, 2$
B	$2, 1$	$4, 0$

Learning and Teaching

Another example: Who will be the teacher?

	Left	Right
Left	1, 1	0, 0
Right	0, 0	1, 1

Repeated games

A one-shot game (stage game) is played repeatedly.

- Do the agents see what others play?
- Do agents have memory?
- What is the utility of the repeated game?
- Is it repeated a finite number of times or infinitely often?

Repeated games

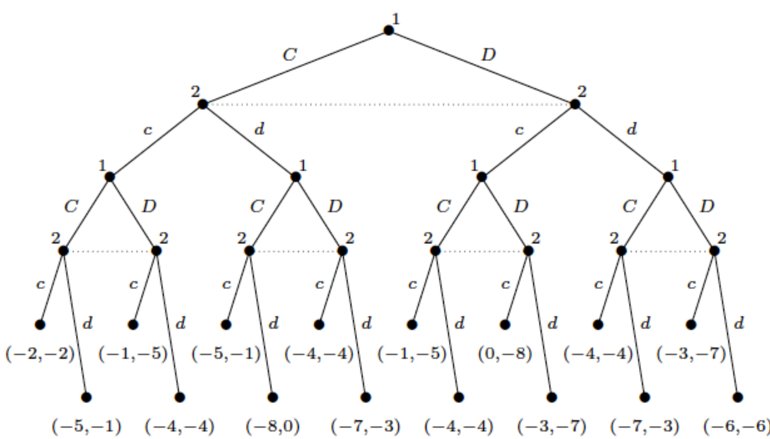
Example: a twice played Prisoners' Dilemma in extensive form (payoff is additive and no player knows what the other plays at that time).

- Stationary strategy
- Backwards induction

for finite repetitions

- What about

infinite repetitions?



Repeated games

Example: repeated Prisoners' Dilemma

What about infinite repetitions?

Definition **(Average reward)** *Given an infinite sequence of payoffs* $r_i^{(1)}, r_i^{(2)}, \ldots$ *for player* i*, the* average reward *of* i *is*

$$\lim_{k\to\infty} \frac{\sum_{j=1}^{k} r_i^{(j)}}{k}.$$

Definition **(Discounted reward)** *Given an infinite sequence of payoffs* $r_i^{(1)}, r_i^{(2)}, \ldots$ *for player* i*, and a discount factor* β *with* $0 \leq \beta \leq 1$*, the* future discounted reward *of* i *is* $\sum_{j=1}^{\infty} \beta^j r_i^{(j)}$.

Repeated games

Example: repeated Prisoners' Dilemma
What about infinite repetitions?

- Other strategies except stationary ones?
  - Tit for Tat (TfT)
  - Trigger Strategy
  - More complex strategies (based on history)
  - Nash equilibria

Stochastic games

Stochastic games are very broad framework, generalizing both Markov decision processes (MDPs) and repeated games.

An MDP is simply a stochastic game with only one player, while a repeated game is a stochastic game in which there is only one stage game.

Stochastic games

Definition **(Stochastic game)** *A stochastic game (also known as a* Markov game*) is a tuple* (Q, N, A, P, r)*, where:*

- Q *is a finite set of games;*
- N *is a finite set of* n *players;*
- $A = A_1 \times \cdots \times A_n$*, where* A_i *is a finite set of actions available to player* i*;*
- $P : Q \times A \times Q \mapsto [0, 1]$ *is the transition probability function;* $P(q, a, \hat{q})$ *is the probability of transitioning from state* q *to state* $\hat{q}$ *after action profile* a*; and*
- $R = r_1, \ldots, r_n$*, where* $r_i : Q \times A \mapsto \mathbb{R}$ *is a real-valued payoff function for player* i*.*

Definition **(Behavioral strategy)** *A behavioral strategy* $s_i(h_t, a_{i_j})$ *returns the probability of playing action* a_{i_j} *for history* h_t*.*

Definition **(Markov strategy)** *A* Markov strategy s_i *is a behavioral strategy in which* $s_i(h_t, a_{i_j}) = s_i(h'_t, a_{i_j})$ *if* $q_t = q'_t$*, where* q_t *and* q'_t *are the final states of* h_t *and* h'_t*, respectively.*

Definition **(Stationary strategy)** *A* stationary strategy s_i *is a Markov strategy in which* $s_i(h_{t_1}, a_{i_j}) = s_i(h'_{t_2}, a_{i_j})$ *if* $q_{t_1} = q'_{t_2}$*, where* q_{t_1} *and* q'_{t_2} *are the final states of* h_{t_1} *and* h'_{t_2}*, respectively.*

What is given (and what is learnt)?

- *The game is given*
  - *Learn the strategies employed by others*

- *The game is not given*
  - *Learn the structure of the game*
  - *Learn the strategies of the others*

- *Other cases can be considered: Learning in large agent societies (scalability and/or emergent behavior).*

- *Learning strategy*

What is given (and what is learnt)?

- *The game is given*
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  - *Learn the strategies of the others*

- *Other cases can be considered: Learning in large agent societies (scalability and/or emergent behavior).*

- *Learning strategy : Complete strategies for learning to choose actions and update agents' beliefs (e.g. what about accumulating knowledge?)*

In-seperability between teaching and learning

Example: Chicken game

What would be your choice to teach the other?

	Yield	Dare
Yield	$2, 2$	$1, 3$
Dare	$3, 1$	$0, 0$

Descriptive and Prescriptive Learning Theories

Descriptive Theories: Build models of real-life learning and prove properties of it.

Properties:

- **Realism**: There should be a good match between the model and the natural phenomenon
- **Convergence**: The strategy profile played converges to some solution concept of the game.
  - *Holy grail: show convergence to stationary strategies that form Nash equilibrium of the stage game.*
  - *Alternatively, show that the empirical frequency of play converge to a Nash Equilibrium*
  - *Or, seek convergence to a correlated equilibrium of the stage game*
  - *Or finally, converge to an interesting state (e.g. correct models of opponents strategies)*

Descriptive and Prescriptive Learning Theories

Prescriptive Theories: Build models of how one **should** learn.

- Achieving optimal learning behaviour depends on the behaviour of the other agents in the system.
- In self-play all agents adopt the same learning strategy.
- Self-play is interesting but no learning procedure is optimal against all possible opponent behaviors.
- What about learning strategies that are in equilibrium with each other (e.g. TfT)?
- More practical: Use learning strategies that are “high enough”
    - Beware: This is neither stronger nor weaker than best response

Descriptive and Prescriptive Learning Theories

Prescriptive Theories: Build models of how one **should** learn.

High-payoff requirements' properties:

- **Safety**: Guarantees at least its maximin payoff, or "security value"
- **Rationality**: The agent settles on the best response strategy to the stationary strategy of the stage game played by the opponent.
- **No-regret (Hannan/universal consistency, informal)**: The learnt strategy played against any set of opponents yields a payoff that is no less than the payoff the agent could have obtained by playing any one of its pure strategies throughout

➢What about relaxing a bit?

Learning Rules (or methods)

- **Fictitious play**: maintaining beliefs about opponents' strategy.
- **Rational (Bayesian) learning**: Enriched beliefs about opponents' strategies (e.g. repeated game strategies not repeated stage-game strategies).
- **Reinforcement (Q-) learning**: Learning without modeling explicitly opponents' strategy.
- **No-regret learning and universal consistency**: Learning a consistent strategy in terms of guaranteeing no less that playing any pure strategy throughout, against any set of opponents.
- **Targeted learning:** targeting good rewards against specific (types) of opponents and self-play, and maxmin against any others.
- **Evolutionary learning and other large-population models**: Study the change in the constitution and behavior of the population over time.

Learning Rules (or methods)

Fictitious play: maintaining beliefs about opponents' strategy (model-based learning)

- to compute Nash equilibria in zero-sum games

Initialize beliefs about the opponent's strategy
repeat
  Play a best response to the assessed strategy of the opponent
  Observe the opponent's actual play and update beliefs accordingly

Notes:

- Think of mixed strategies: The agent believes that the opponent plays the mixed strategy given by the empirical distribution of opponents' previous actions.
- Agent is oblivious to the payoffs of others
- The agent is assumed to know own payoffs in the stage game

Learning Rules (or methods)

Fictitious play

Initialize beliefs about the opponent's strategy
repeat
    Play a best response to the assessed strategy of the opponent
    Observe the opponent's actual play and update beliefs accordingly

The agent believes that the opponent plays the mixed strategy given by the empirical distribution of opponents' previous actions.

- For every action a in opponents set A , it maintains $w(a)$: The number of times a has been played.

The **empirical distribution** is provided by

$$P(a) = \frac{w(a)}{\sum_{a' \in A} w(a')}.$$

- Tie-breaking (best response) rule: no particular effect on the results of the learning rule.

Learning Rules (or methods)

Fictitious play

Initialize beliefs about the opponent's strategy
repeat
  Play a best response to the assessed strategy of the opponent
  Observe the opponent's actual play and update beliefs accordingly

The agent believes that the opponent plays the mixed strategy given by the empirical distribution of opponents' previous actions.

- Very sensitive to initialization of beliefs
- Assumes a stationary strategy of the opponent (what happened to teaching and learning?)

Learning Rules (or methods)

Fictitious play

Initialize beliefs about the opponent's strategy
repeat
  Play a best response to the assessed strategy of the opponent
  Observe the opponent's actual play and update beliefs accordingly

	Heads	Tails
Heads	$1, -1$	$-1, 1$
Tails	$-1, 1$	$1, -1$

Round	1's action	2's action	1's beliefs	2's beliefs
0			(1.5,2)	(2,1.5)
1	T	T	(1.5,3)	(2,2.5)
2	T	H	(2.5,3)	(2,3.5)
3	T	H	(3.5,3)	(2,4.5)
4	H	H	(4.5,3)	(3,4.5)
5	H	H	(5.5,3)	(4,4.5)
6	H	H	(6.5,3)	(5,4.5)
7	H	T	(6.5,4)	(6,4.5)
⋮	⋮	⋮	⋮	⋮

It converges to (0.5,0.5): the unique Nash equilibrium of the normal form stage game.

Learning Rules (or methods)

Fictitious play

Simple, but with nice properties:

Definition (Steady state) *An action profile a is a* steady state *(or* absorbing state*) of fictitious play if it is the case that whenever a is played at round t it is also played at round $t+1$ (and hence in all future rounds as well).*

Theorem *If a pure-strategy profile is a strict Nash equilibrium of a stage game, then it is a steady state of fictitious play in the repeated game.*

Theorem *If a pure-strategy profile is a steady state of fictitious play in the repeated game, then it is a (possibly weak) Nash equilibrium in the stage game.*

While the stage game strategies may not converge, the empirical distribution of the the stage game strategies over multiple iterations may (as e.g. in the matching pennies case)

Theorem *If the empirical distribution of each player's strategies converges in fictitious play, then it converges to a Nash equilibrium.*

Learning Rules (or methods)

Fictitious play

Theorem *If the empirical distribution of each player's strategies converges in fictitious play, then it converges to a Nash equilibrium.*

No claims about the distribution of the particular outcomes played is made

	A	B
A	0, 0	1, 1
B	1, 1	0, 0

Round	1's action	2's action	1's beliefs	2's beliefs
0			(1,0.5)	(1,0.5)
1	B	B	(1,1.5)	(1,1.5)
2	A	A	(2,1.5)	(2,1.5)
3	B	B	(2,2.5)	(2,2.5)
4	A	A	(3,2.5)	(3,2.5)
⋮	⋮	⋮	⋮	⋮

*Players converge to the mixed strategy Nash equilibrium, but they may **<u>not</u>** receive the expected payoff of the Nash equilibrium, due to miscorrelated actions.*

Finally, no claims about the final convergence are made.

Learning Rules (or methods)

Fictitious play

Theorem *Each of the following is a sufficient condition for the empirical frequencies of play to converge in fictitious play:*

- *The game is zero sum;*
- *The game is solvable by iterated elimination of strictly dominated strategies;*
- *The game is a potential game;*
- *The game is* $2 \times n$ *and has generic payoffs.*

Conclusion: Simple leaning rule, but mathematically constrained and unplausible for modelling human learning: What about smooth fictitious play?

Learning Rules (or methods)

Rational (Bayesian) Learning

Model – based learning, as done in Fictitious play.

However:

- The set of strategies of the opponent can include repeated game strategies (e.g. TfT)
- The beliefs may be expressed by any probability distribution over the set of all possible strategies

Learning Rules (or methods)

Rational (Bayesian) Learning

Agents start with prior beliefs for opponents

At each round they update beliefs using Bayesian updating.

- S^i_{-i} denote the set of opponent strategies considered by i

- H denote the possible histories of the game.

> *Given a particular strategy* s-i *in* S^i_{-i} *and a particular history* h *the updating rule is*

$$P_i(s_{-i}|h) = \frac{P_i(h|s_{-i})P_i(s_{-i})}{\sum_{s'_{-i} \in S^i_{-i}} P_i(h|s'_{-i})P_i(s'_{-i})}.$$

Learning Rules (or methods)

Rational (Bayesian) Learning

Example:

$$P_i(s_{-i}|h) = \frac{P_i(h|s_{-i})P_i(s_{-i})}{\sum_{s'_{-i}\in S^i_{-i}} P_i(h|s'_{-i})P_i(s'_{-i})}.$$

	C	D
C	$3,3$	$0,4$
D	$4,0$	$1,1$

Suppose that the support set of each player consists of the strategies

$$g_1, g_2, \cdots g_\infty$$

Where the last is the *trigger strategy, but for any other T , is the trigger strategy until t<T and then defection.*

Also the agents play the game with a strategy profile

$$(g_{T_1}, g_{T_2})$$

Learning Rules (or methods)

Rational (Bayesian) Learning

Very involved formal analysis, focusing mostly on self-play.

$$P_i(s_{-i}|h) = \frac{P_i(h|s_{-i})P_i(s_{-i})}{\sum_{s'_{-i} \in S^i_{-i}} P_i(h|s'_{-i})P_i(s'_{-i})}.$$

Highlights

- Under some conditions, in *self-play* rational learning results in agents having close to correct beliefs about the observable portion of their opponent's strategy
- Under some conditions, in *self-play* rational learning causes the agents to converge toward a Nash equilibrium with high probability
- Chief among these "conditions" is **absolute continuity**, a <u>**strong**</u> assumption

Learning Rules (or methods)

Rational (Bayesian) Learning

Definition (Absolute continuity) *Let* X *be a set and let* $\mu, \mu' \in \Pi(X)$ *be probability distributions over* X. *Then the distribution* μ *is said to be* absolutely continuous *with respect to the distribution* μ' *iff for* $x \subset X$ *that is measurable it is the case that if* $\mu(x) > 0$ *then* $\mu'(x) > 0$.

Learning Rules (or methods)

Rational (Bayesian) Learning

Beliefs, history and absolute continuity

- Given a strategy profile, we can induce a probability distribution on H
- Given player's beliefs on opponents, one can also deduce such a distribution

- Let S^i_j be a set of strategies that i believes possible for j, and $P^i_j \in \Pi(S^i_j)$ be the distribution over S^i_j believed by player i.

  $P^i = (P^i_1, P^i_2, \ldots P^i_n)$: Beliefs of player i for the strategies of players.

Using these beliefs, the agent can induce a probability distribution over H .

- So, results that follow require that the distribution over histories induced by the actual strategies is **absolutely continuous** with respect to the distribution induced by a player's beliefs

Learning Rules (or methods)

Rational (Bayesian) Learning

Analysis results (overall)

Players satisfying the assumptions of the rational learning model will have beliefs about the play of the other players that converge to the truth, and furthermore, players will in finite time converge to play that is arbitrarily close to the Nash equilibrium.

Learning Rules (or methods)

Rational (Bayesian) Learning

Definition **(ϵ-closeness)** *Given an $\epsilon > 0$ and two probability measures μ and μ' on the same space, we say that μ is ϵ-close to μ' if there is a measurable set Q satisfying:*

- *$\mu(Q)$ and $\mu'(Q)$ are each greater than $1 - \epsilon$; and*
- *For every measurable set $A \subseteq Q$, we have that*

$$(1+\epsilon)\mu'(A) \geq \mu(A) \geq (1-\epsilon)\mu'(A).$$

Learning Rules (or methods)

Rational (Bayesian) Learning

Theorem (Rational learning and belief accuracy) *Let s be a repeated-game strategy profile for a given n-player game , and let $P = P_1, \ldots, P_n$ be a tuple of probability distributions over such strategy profiles (P_i is interpreted as player i's beliefs). Let μ_s and μ_P be the distributions over infinite game histories induced by the strategy profile s and the belief tuple P, respectively. If we have that*

- *at each round, each player i plays a best response strategy given his beliefs P_i;*
- *after each round each player i updates P_i using Bayesian updating; and*
- *μ_s is absolutely continuous with respect to μ_{P_i},*

then for every $\epsilon > 0$ and for almost every history in the support of μ_s (i.e., every possible history given the actual strategy profile s), there is a time T such that for all $t \geq T$, the play μ_{P_i} predicted by the player i's beliefs is ϵ-close to the distribution of play μ_s predicted by the actual strategies.

Note that this result does not state that players will learn the true strategy being played by their opponents.

Instead, players' beliefs will accurately predict the play of the game,
and no claim is made about their accuracy in predicting the off-path portions of the opponents' strategies.

Learning Rules (or methods)

Rational (Bayesian) Learning

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Instead, players' beliefs will accurately predict the play of the game, and no claim is made about their accuracy in predicting the off-path portions of the opponents' strategies.

Examples in Prisoners' Dilemma infinitely repeated game:

- *Let a profile with* (g_{T1} , g_{T2}), T1<T2

- Tit-for-Tat against g_{inf}

Learning Rules (or methods)

Rational (Bayesian) Learning

Theorem ' (Rational Learning and Nash) *Let s be a repeated-game strategy profile for a given n-player game, and let $P = P_1, \ldots, P_n$ be a a tuple of probability distributions over such strategy profiles. Let μ_s and μ_P be the distributions over infinite game histories induced by the strategy profile s and the belief tuple P, respectively. If we have that*

- *at each round, each player i plays a best response strategy given his beliefs P_i;*
- *after each round each player i updates P_i using Bayesian updating; and*
- *μ_s is absolutely continuous with respect to μ_{P_i},*

then for every $\epsilon > 0$ and for almost every history in the support of μ_s there is a time T such that for every $t \geq T$ there exists an ϵ-equilibrium s^* of the repeated game in which the play μ_{P_i} predicted by player i's beliefs is ϵ-close to the play μ_{s^*} of the equilibrium.

Learning Rules (or methods)

Reinforcement Learning

Learning in unknown MDPs (Q-learning)

It turns out that if the agent knows the state is in and the reward received, then it converges to the correct Q-values

Extended to multiagent settings

Learning in unknown MDPs (Q-learning)

Definition (Q**-learning**) *Q-learning is the following procedure:*

Initialize the Q-function and V values (arbitrarily, for example)
repeat *until convergence*
- Observe the current state s_t.
- Select action a_t and and take it.
- Observe the reward $r(s_t, a_t)$
- Perform the following updates (and do not update any other Q-values):
- $Q_{t+1}(s_t, a_t) \leftarrow (1 - \alpha)Q_t(s_t, a_t) + \alpha_t(r(s_t, a_t) + \beta V_t(s_{t+1}))$
- $V_{t+1}(s) \leftarrow \max_a Q_t(s, a)$

Learning Rules (or methods)

Reinforcement Learning

Learning in unknown MDPs (Q-learning)

Theorem *Q-learning guarantees that the Q and V values converge to those of the optimal policy, provided that each state-action pair is sampled an infinite number of times, and that the time-dependent learning rate α_t obeys $0 \leq \alpha_t < 1$, $\sum_0^\infty \alpha_t = \infty$ and $\sum_0^\infty \alpha_t^2 < \infty$.*

Does not guarantee quick learning nor high future discounted reward (if not in a simulator)

Learning Rules (or methods)

Reinforcement Learning

Learning in unknown MDPs (Q-learning)

Adopt an individual learners' scheme: Each one computes own Q-values, given the common state and own action.

$$Q_i^\pi : S \times A_i \mapsto \mathbb{R}$$

The multiagent setting forces us to forego the optimal strategy and instead focus on one that performs well against the opponent.

Q-learning can be shown to be Hannan -consistent for an agent in a stochastic game against opponents playing stationary policies.
However, against opponents using more complex strategies, such as Q-learning itself, we do not obtain such a guarantee.

Learning Rules (or methods)

Reinforcement Learning

Learning in unknown MDPs (Q-learning)

What if the agent knows the actions of the opponent?

*Then the Q value is computed over **joined** states and action profiles*

$$Q_i^{\pi} : S \times A \mapsto \mathbb{R}$$

The Q update formula is as follows:

$$Q_{i,t+1}(s_t, a_t, o_t) = (1 - \alpha_t) Q_{i,t}(s_t, a_t, o_t) + \alpha_t (r_i(s_t, a_t, o_t) + \beta V_t(s_{t+1}))$$

How can we calculate the value for a state?
Recall that for (two-player) zero-sum games, the policy profile where each agent plays its maxmin strategy forms a Nash equilibrium. The payoff to the first agent (and thus the negative of the payoff to the second agent) is called the value of the game. Thus,

$$V_t(s) = \max_{\Pi_i} \min_{o} Q_{i,t}(s, \Pi_i(s), o).$$

Learning Rules (or methods)

Reinforcement Learning

Learning in unknown MDPs (Q-learning)

Minmax Q learning

$$Q_i^\pi : S \times A \mapsto \mathbb{R}$$

$$Q_{i,t+1}(s_t, a_t, o_t) = (1 - \alpha_t) Q_{i,t}(s_t, a_t, o_t) + \alpha_t (r_i(s_t, a_t, o_t) + \beta V_t(s_{t+1}))$$

$$V_t(s) = \max_{\Pi_i} \min_o Q_{i,t}(s, \Pi_i(s), o).$$

While this will guarantee the agent a payoff at least equal to that of its maxmin strategy, it no longer satisfies Hannan consistency.

If the opponent is playing a suboptimal strategy, minimax-Q will be unable to exploit it in most games.

Learning Rules (or methods)

Reinforcement Learning

Learning in unknown MDPs (Q-learning)

Minmax Q learning

// Initialize:

forall $s \in S, a \in A,$ *and* $o \in O$ **do**

$\quad Q(s,a,o) \leftarrow 1$

forall s *in* S **do**

$\quad V(s) \leftarrow 1$

forall $s \in S$ *and* $a \in A$ **do**

$\quad \Pi(s,a) \leftarrow 1/|A|$

$\alpha \leftarrow 1.0$

// Take an action:

when in state s, with probability *explor* choose an action uniformly at random, and with probability $(1 - explor)$ choose action a with probability $\Pi(s,a)$

// Learn:

after receiving reward rew for moving from state s to s' via action a and opponent's action o

$Q(s,a,o) \leftarrow (1-\alpha) * Q(s,a,o) + \alpha * (rew + \gamma * V(s'))$

$\Pi(s,\cdot) \leftarrow \arg\max_{\Pi'(s,\cdot)}(\min_{o'} \sum_{a'} (\Pi(s,a') * Q(s,a',o')))$

// The above can be done, for example, by linear programming

$V(s) \leftarrow \min_{o'}(\sum_{a'} (\Pi(s,a') * Q(s,a',o')))$

Update α

Learning Rules (or methods)

Reinforcement Learning

Learning in unknown MDPs (Q-learning)

Theorem *Under the same conditions that assure convergence of Q-learning to the optimal policy in MDPs, in zero-sum games Minimax-Q converges to the value of the game in self play.*

Expanding reinforcement learning algorithms to the general-sum case is quite problematic. There have been attempts to generalize Q-learning to general-sum games, but they have not yet been truly successful.
The question of what it means to learn in general-sum games is subtle. One yardstick we have discussed is convergence to Nash equilibrium of the stage game during self play.
No generalization of Q-learning has been put forward that has this property.

Learning Rules (or methods)

Reinforcement Learning

Learning in unknown MDPs (Q-learning)

Belief based Q learning

$$Q_i^\pi : S \times A \mapsto \mathbb{R}$$

$$Q_{t+1}(s_t, a_t) \leftarrow (1-\alpha)Q_t(s_t, a_t) + \alpha_t(r(s_t, a_t) + \beta V_t(s_{t+1}))$$

$$V_t(s) \leftarrow \max_{a_i} \sum_{a_{-i} \subset A_{-i}} Q_t(s, (a_i, a_{-i})) Pr_i(a_{-i})$$

There are some experimental results that show convergence to equilibrium in self-play for some versions of belief-based reinforcement learning and some classes of games, but no theoretical results.